

TRIGGER

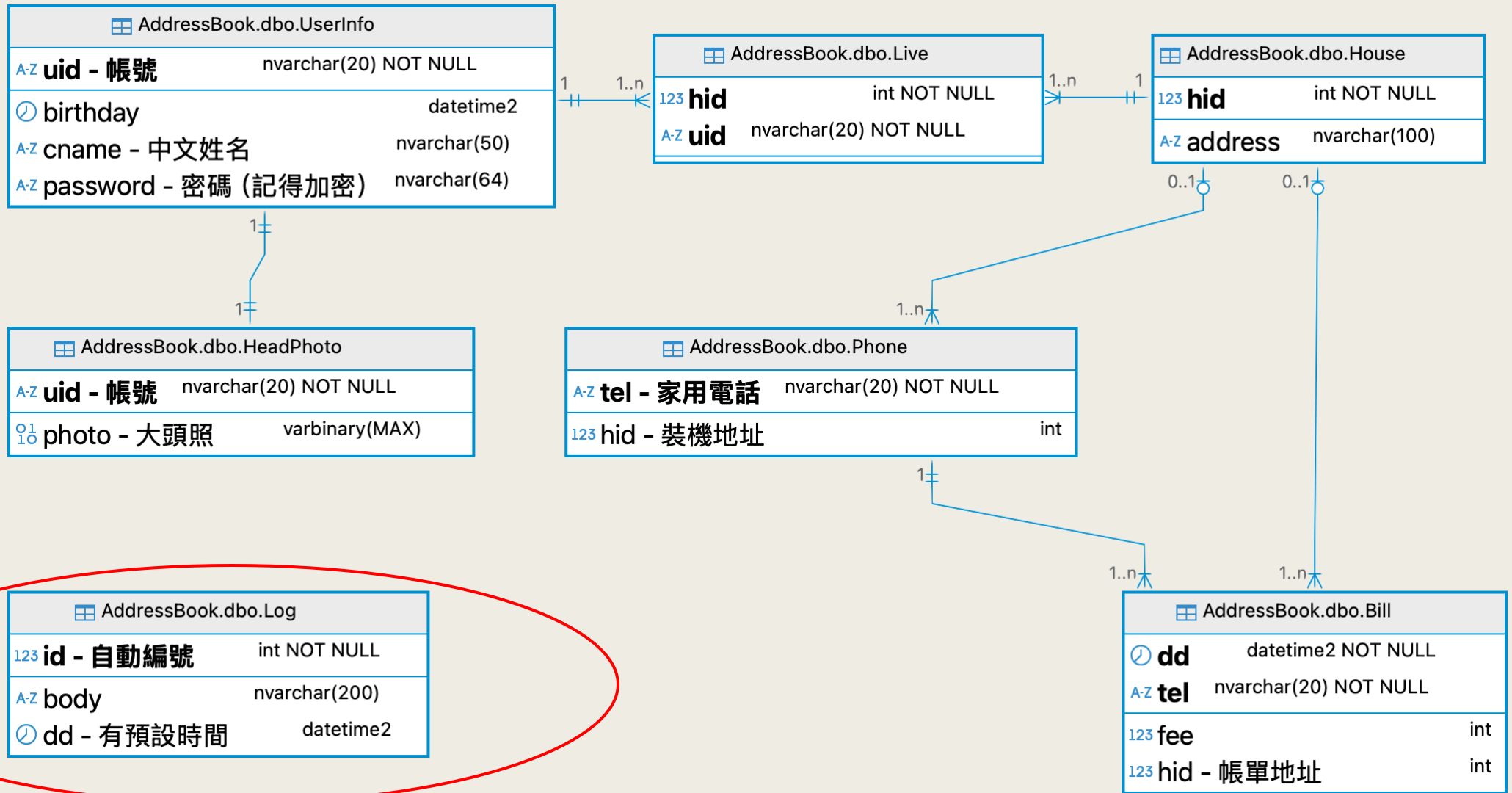
朱克剛



用途

- 讓資料異動時可以連鎖觸發另一個資料表異動





自動寫入 Log

- 當 UserInfo 新增資料時，自動在 Log 資料表中記錄這筆異動

```
create trigger trg_InsertUserInfo
on UserInfo
after insert
as
begin
    insert into Log (body)
    select concat(
        '新增使用者:', uid,
        ', 姓名:', isnull(cname, '未輸入')
    )
    from inserted
end
```

三合一的 trigger

- UserInfo 資料表只要有異動，這個 trigger 都會被觸發

```
create trigger trg_AllModifyUserInfo
on UserInfo
after insert, update, delete
as
begin
    if exists (select 1 from inserted) and exists (select 1 from deleted)
    begin
        -- UPDATE
    end
    else if exists (select 1 from inserted)
    begin
        -- INSERT
    end
    else if exists (select 1 from deleted)
    begin
        -- DELETE
    end
end
```

判斷是否修改了某個欄位 – update()

- 如果修改密碼欄位，不可以一次修改兩筆以上

```
create trigger trg_UpdateUserInfoPassword
on UserInfo
after update
as
begin
    if update(password) and (select count(*) from inserted) > 1
    begin
        rollback
        raiserror('不可一次修改兩筆以上密碼', 16, 1)
    end
end
```

異動前觸發 – Instead Of trigger

- 使用 trigger 檔掉一次刪除兩筆以上資料

```
create trigger trg_InsteadOfDeleteUserInfo  
on UserInfo  
instead of delete  
as  
begin  
  if (select count(*) from deleted) >= 2  
    begin  
      raiserror('不可一次刪除兩筆以上資料', 16, 1)  
    end  
  else  
    begin  
      delete from UserInfo where uid in (select uid from deleted)  
    end  
end
```

INSERT
UPDATE
DELETE



觸
發
器

資料表

Trigger 管理

- 查詢有哪些 Trigger

```
select object_name(parent_id) as TableName, *  
from sys.triggers
```

- 停用 Trigger (all 可以改為 trigger 名字)

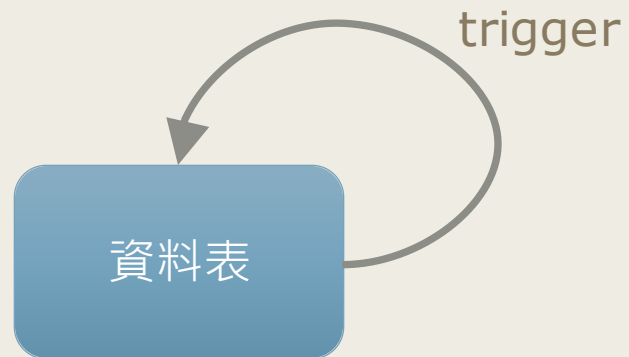
```
disable trigger all on UserInfo
```

- 啟用 Trigger

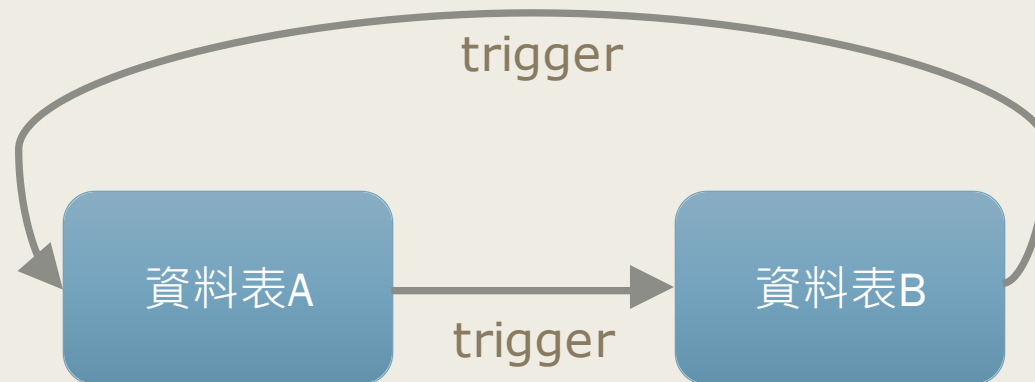
```
enable trigger all on UserInfo
```


遞迴 (recursive)

- 直接遞迴



- 間接遞迴



遞迴偵測

- 遞迴層級最大可達 32 層
 - 透過「重複觸發程序」選項開啟「直接遞迴」功能
 - 該資料庫 → 滑鼠右鍵 → 屬性 → 選項
- 該選項只能關閉「直接遞迴」
- 在遞迴中只要執行到 rollback 指令，遞迴結束